

Mathematics Bootcamp

General Equilibrium and the Social Planner

Miguel Vázquez-Carrero

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Today

- Infinite Paths and Boundary Conditions
 - Linking choices, recurrences, and the condition at infinity
- The Household's Dynamic Optimization Problem
 - Dated goods, shadow values, and the Euler equation
- Competitive Equilibrium
 - Definition, market clearing, and characterization
- The Planner as a Concluding Application
 - Efficiency, supporting prices, and existence

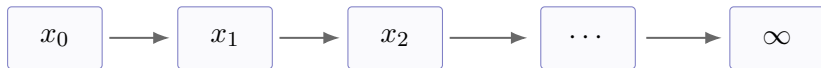
Day 4 Chose a Point; Day 5 Chooses an Infinite Path

Day 4 studied a point

$$x \in \mathbb{R}^n.$$

Today the choice is an infinite sequence:

$$z = (x_0, x_1, x_2, \dots).$$



An infinite-horizon problem has one local condition at every date and one boundary condition at infinity.

Every Date Contributes a Constraint and a Multiplier

Consider an infinite-path problem and its formal Lagrangean:

$$\max_{z=(z_t)_{t=0}^{\infty}} F(z) \quad \text{s.t.} \quad g_t(z) = 0, t = 0, 1, \dots, \quad L(z; \lambda) = F(z) + \sum_{t=0}^{\infty} \lambda_t g_t(z).$$

$$g_0(z) = 0$$

$$\lambda_0$$

$$g_1(z) = 0$$

$$\lambda_1$$

...

...

$$g_t(z) = 0$$

$$\lambda_t$$

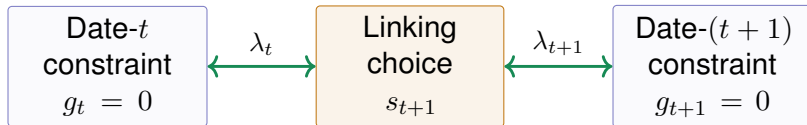
Regularity and Interpretation

Assume an infinite-horizon multiplier theorem supplies λ . Under its regularity conditions, the multiplier sequence is unique. For $g_t(z) + b_t = 0$ and differentiable V , $\lambda_t = \partial V(b) / \partial b_t$.

A Linking Choice Creates a Recurrence

A variable carried from t to $t + 1$ appears in two neighboring constraints:

$$g_t(\dots, s_{t+1}) = 0, \quad g_{t+1}(s_{t+1}, \dots) = 0.$$



Differentiating with respect to s_{t+1} links adjacent multipliers. Eliminating them typically produces

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0.$$

This is a second-order difference equation.

An Infinite Path Needs a Boundary at Infinity

A recurrence alone does not determine a path:

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0, \quad t = 0, 1, \dots$$

It needs an initial condition and a boundary condition at infinity:

$$s_0 \text{ given,} \quad \lim_{t \rightarrow \infty} \lambda_t s_{t+1} = 0.$$

Infinite-Horizon Logic

- Existence requires an argument about feasible sequences and tails.
- Strict concavity gives at most one solution.
- Regularity gives first-order conditions at every date.
- Concavity and the boundary condition verify global optimality.

Time for Economics

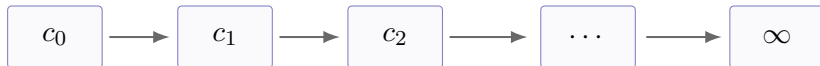
- The vector z becomes a path of consumption, assets, and capital.
- Each g_t becomes a dated budget or resource constraint.
- The linking choice s_{t+1} becomes saving or next-period capital.
- When the perturbed value is differentiable, λ_t becomes the shadow value of resources at date t .
- The recurrence becomes an Euler equation.

The Same Mathematical Structure

Dynamic economics applies ordinary constrained optimization to choices that are linked across dates.

Consumption at Different Dates Means Different Commodities

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label:



Different Date, Different Good

Consumption at t and $t + 1$ are different goods. Saving exchanges one for the other.

The Interest Rate Is a Relative Price

One unit at date $t + 1$ costs $1/(1 + r_{t+1})$ units at date t . Trades occur date by date.

$$\text{relative price of } c_{t+1} = \frac{1}{1 + r_{t+1}}.$$

The Environment Is a List of Primitives

Dynamic Production Economy

The environment is

$$E = (\mathbb{N}_0, \beta, u, f, (A_t)_{t=0}^{\infty}, k_0) .$$

These primitives are fixed before prices and allocations are determined.

Primitives

- Dates $t = 0, 1, \dots$
- Preferences: β and u
- Technology: f and A_t

Initial Resources

- The household owns $a_0 = k_0 > 0$.
- Labor supply is one unit each date.
- Capital does not depreciate: $\delta = 0$.

Infinite Horizon Adds a Boundary at Infinity

The model uses dates $t = 0, 1, \dots$ and assumes:

- a deterministic closed economy with no government;
- one representative price-taking household and firm;
- u is strictly increasing, strictly concave, and twice differentiable; f is increasing, concave, and constant returns to scale;
- $k_0 = a_0 > 0$ and $\delta = 0$.

Transversality Condition

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) a_{t+1} = 0.$$

The Euler equation is local; this condition controls the tail.

The Household Ranks Entire Consumption Paths

Lifetime Utility

For $c = (c_0, c_1, c_2, \dots)$,

$$U(c) = \sum_{t=0}^{\infty} \beta^t u(c_t), \quad 0 < \beta < 1.$$

The discount factor β gives the relative weight on later utility.

Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

Strict concavity

$$u''(c_t) < 0$$

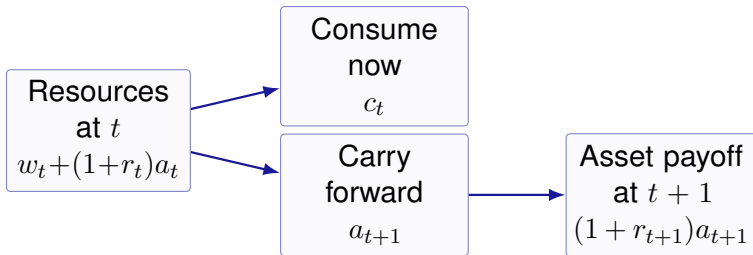
Marginal utility falls with consumption.

The Household Transfers Resources Across Dates

At date t , the household owns assets a_t , supplies one unit of labor, and receives wage and capital income.

Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$



The Household Chooses an Infinite Path at Given Prices

Given the price paths $(w_t, r_t)_{t=0}^{\infty}$ and initial assets a_0 , the household solves

$$\max_{(c_t, a_{t+1})_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \quad c_t \geq 0, \quad a_{t+1} \geq 0, \quad t = 0, 1, \dots,$$

a_0 given.

Admissibility and Price Taking

Nonnegative assets rule out Ponzi finance. The household chooses quantities while treating every wage and interest rate as given.

One Shadow Value Is Needed for Every Date

At an interior path, write the formal Lagrangean

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [w_t + (1 + r_t)a_t - c_t - a_{t+1}].$$

The first-order conditions imply

$$\beta^t u'(c_t) = \lambda_t, \quad t = 0, 1, \dots,$$

$$\lambda_t = (1 + r_{t+1})\lambda_{t+1}, \quad t = 0, 1, \dots$$

Interpretation

λ_t is the increase in lifetime utility from one additional unit of the consumption good available at date t .

The Euler Equation Balances Sacrifice and Return

Combine the two first-order conditions:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

Benefit One Date Later

Tomorrow it yields $1 + r_{t+1}$ units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

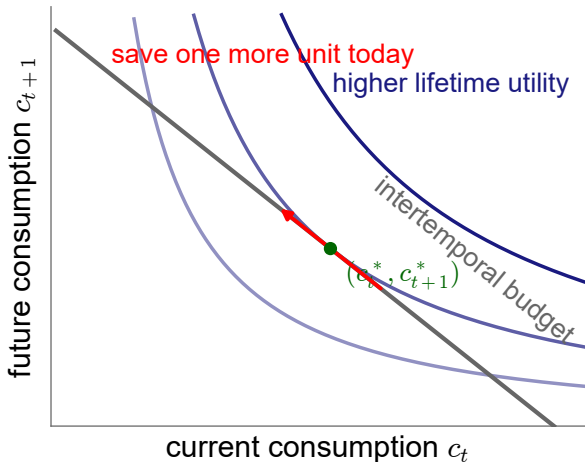
The Euler Equation Is Ordinary Tangency Across Dates

The household treats consumption at t and $t + 1$ as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1 + r_{t+1}}.$$

Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



The Firm Transforms Capital and Labor into Output

In every date t , the firm rents capital k_t , hires labor ℓ_t , and produces the single final good:

$$y_t = A_t f(k_t, \ell_t) = A_t k_t^\alpha \ell_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

Technology

A_t is total factor productivity. It shifts output without changing the capital–labor ratio inside f .

Constant Returns

Scaling both inputs by one factor scales output by that factor. Price taking then implies zero profits.

Throughout, $\delta = 0$: capital is fully preserved between dates.

The Firm Solves a Static Problem at Every Date

Given the wage w_t and rental rate r_t , the firm solves

$$\max_{k_t, \ell_t \geq 0} \{A_t f(k_t, \ell_t) - w_t \ell_t - r_t k_t\}.$$

At a positive-input optimum,

$$w_t = A_t \frac{\partial f(k_t, \ell_t)}{\partial \ell_t} = A_t f_2(k_t, \ell_t), \quad r_t = A_t \frac{\partial f(k_t, \ell_t)}{\partial k_t} = A_t f_1(k_t, \ell_t).$$

Why the Firm Problem Is Static

This is the one-period problem from Day 4, repeated at every date. The firm rents inputs, produces, and sells output within date t ; the household owns the capital and chooses how much to carry into the next date.

The Same Rental Rate Links Firm and Household

For Cobb–Douglas production,

$$w_t = (1 - \alpha)A_t k_t^\alpha \ell_t^{-\alpha}, \quad r_t = \alpha A_t k_t^{\alpha-1} \ell_t^{1-\alpha}.$$

The firm pays r_t for one unit of capital. Since capital is returned intact, the household receives the gross payoff $1 + r_t$.

Constant returns and the first-order conditions imply

$$A_t f(k_t, \ell_t) = w_t \ell_t + r_t k_t,$$

so competitive profits are zero.

DEFINITION: Allocation and Feasibility

Allocation

A physical allocation is a sequence

$$z = (c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}.$$

It is **feasible** if, for every $t = 0, 1, \dots$,

$$c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq \ell_t \leq 1,$$

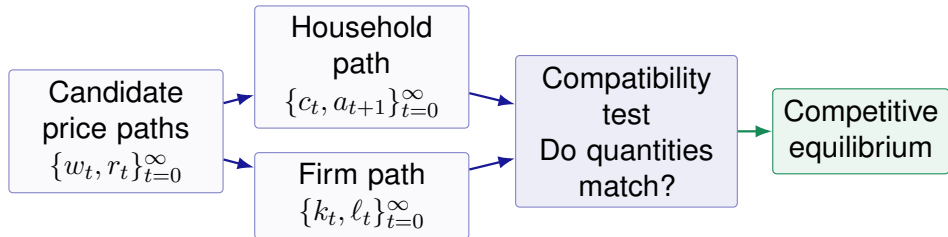
and

$$c_t + k_{t+1} \leq A_t k_t^{\alpha} \ell_t^{1-\alpha} + k_t.$$

Feasibility is physical and contains no prices. Monotonicity makes the resource inequality bind at an optimum.

General Equilibrium Makes Separate Choices Compatible

The household and firm have now solved separate optimization problems. General equilibrium asks whether their choices can hold at the same prices.



The Definition Is a Test, Not an Adjustment Story

It says which price-allocation paths qualify as equilibria. It does not say how an economy discovers those prices.

Optimal Choice May Be Set-Valued

Household Correspondence

At prices (w, r) , define

$$H(w, r) = \{\text{household-optimal paths}\}.$$

It may contain more than one path (c, a) .

Firm Correspondence

At prices (w_t, r_t) , define

$$F_t(w_t, r_t) = \{\text{optimal input pairs at } t\}.$$

It may contain more than one pair (k, ℓ) .

Under constant returns, F_t may contain many input choices. Equilibrium selects elements of both correspondences that are mutually compatible.

Definition and Characterization Answer Different Questions

DEFINITION

- Names the objects that constitute an equilibrium.
- States optimization and market clearing.
- Tests whether a proposed price-allocation path qualifies.
- Does not require derivatives or first-order conditions.

CHARACTERIZATION

- Derives equations every equilibrium must satisfy.
- Uses first-order conditions or KKT under their hypotheses.
- Reduces the objects that must be solved for.
- Needs extra conditions for sufficiency, existence, or uniqueness.

DEFINITION: Competitive Equilibrium

Competitive Equilibrium

Given a_0 , a competitive equilibrium is a collection

$$((c_t, a_{t+1}), (k_t, \ell_t), (w_t, r_t))_{t=0}^{\infty}$$

such that:

1. $(c, a) \in H(w, r)$;
2. $(k_t, \ell_t) \in F_t(w_t, r_t)$ for every t ;
3. markets clear at every date:

$$a_t = k_t, \quad \ell_t = 1, \quad c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

Market Clearing Links Individual and Aggregate Quantities

Asset Market

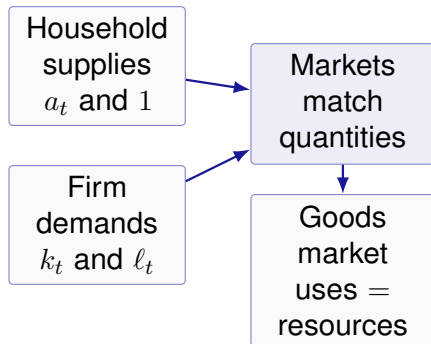
$$a_t = k_t.$$

Labor Market

$$\ell_t = 1.$$

Goods Market

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$



Walras' Law Makes One Clearing Condition Redundant

Suppose the household and firm optimize and the asset and labor markets clear.

(1) Substitute $a_t = k_t$ and $a_{t+1} = k_{t+1}$ into the household budget:

$$c_t + k_{t+1} = w_t + (1 + r_t)k_t.$$

(2) Subtract the capital carried into date t :

$$c_t + k_{t+1} - k_t = w_t + r_t k_t.$$

(3) Since $\ell_t = 1$ and constant returns imply zero profit,

$$w_t \ell_t + r_t k_t = A_t f(k_t, \ell_t).$$

(4) Therefore the goods market clears:

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

CHARACTERIZATION: Prices Become Functions of Capital

To characterize equilibrium, set $A_t = 1$ and use $\ell_t = 1$.

General Technology

$$w_t = f_\ell(k_t, 1),$$

$$r_t = f_k(k_t, 1).$$

Cobb–Douglas Technology

$$w_t = (1 - \alpha)k_t^\alpha,$$

$$r_t = \alpha k_t^{\alpha-1}.$$

Firm optimality and market clearing remove prices as independent unknowns.

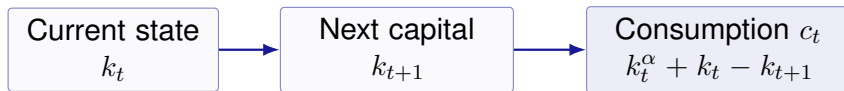
CHARACTERIZATION: Feasibility Eliminates Consumption

With $\ell_t = 1$ and $A_t = 1$, the resource constraint is

$$c_t + k_{t+1} = k_t^\alpha + k_t.$$

Hence,

$$c_t = k_t^\alpha + k_t - k_{t+1}.$$



Once the capital path is known, the allocation and prices can be recovered.

CHARACTERIZATION: A Difference Equation for Capital

For $t = 0, 1, \dots$, substitute equilibrium prices and consumption into the household Euler equation:

$$\begin{aligned} u'(k_t^\alpha + k_t - k_{t+1}) \\ = \beta [1 + \alpha k_{t+1}^{\alpha-1}] u'(k_{t+1}^\alpha + k_{t+1} - k_{t+2}). \end{aligned}$$

This is a second-order difference equation in (k_t, k_{t+1}, k_{t+2}) . Combine it with

$$k_0 = a_0, \quad c_t \geq 0, \quad k_t \geq 0, \quad \lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

A Characterization Is Not an Existence Theorem

The capital equation is useful because it removes prices and consumption from the search. But the logical claims remain separate:

- Every interior competitive equilibrium satisfies the equation.
- The converse also requires all feasibility and boundary conditions, together with conditions making the first-order conditions sufficient.
- Writing the equation does not prove that a solution exists.
- It also does not prove that the solution is unique.

The Next Step

The social planner problem supplies a simpler route to existence, allocation uniqueness, and the welfare interpretation.

DEFINITION: Efficiency Uses Only Feasible Allocations

Efficiency

A feasible allocation $z^* = (c_t^*, k_{t+1}^*, \ell_t^*)_{t=0}^{\infty}$ is efficient if there is no feasible $\tilde{z}$ such that $U(\tilde{c}) > U(c^*)$.

Efficiency uses only preferences and physical feasibility: no prices and no ownership rule. With one household, the planner selects the efficient path by maximizing the household's lifetime utility over feasible allocations.

DEFINITION: The Social Planner Problem

With $A_t = 1$, the planner solves

$$\max_{(c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = k_t^{\alpha} \ell_t^{1-\alpha} + k_t, \quad t = 0, 1, \dots,$$

$$c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq \ell_t \leq 1, \quad k_0 > 0 \text{ given.}$$

Since leisure does not enter utility and production increases in labor, $\ell_t = 1$. At an interior optimum,

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

CHARACTERIZATION: The Planner Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [k_t^\alpha + k_t - c_t - k_{t+1}].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \quad t = 0, 1, \dots,$$

$$\lambda_t = \lambda_{t+1} [\alpha k_{t+1}^{\alpha-1} + 1], \quad t = 0, 1, \dots$$

Therefore,

$$\boxed{u'(c_t) = \beta [\alpha k_{t+1}^{\alpha-1} + 1] u'(c_{t+1}).}$$

The Market and Planner Euler Equations Coincide

Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1}.$$

=

Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1}.$$

Planner Euler equation:

$$u'(c_t) = \beta [1 + \alpha k_{t+1}^{\alpha-1}] u'(c_{t+1}).$$

The equality is economically powerful only because concavity and transversality make these equations sufficient.

The Welfare Theorems Connect Two Definitions

First Welfare Theorem

Every equilibrium allocation is a planner optimum:

equilibrium $\implies$ planner optimum.

Markets attain an efficient allocation.

Second Welfare Theorem

In this model, every planner optimum can be supported by prices:

planner optimum $\implies$ equilibrium.

Marginal products construct the prices.

Under the maintained infinite-horizon assumptions, verification uses feasibility, identical Euler equations, transversality, and concavity.

Supporting Prices Decentralize the Planner Allocation

Start from a planner solution $(c_t^*, k_{t+1}^*)_{t=0}^\infty$ and define

$$w_t^* = (1 - \alpha)(k_t^*)^\alpha, \quad r_t^* = \alpha(k_t^*)^{\alpha-1}.$$

- (1) Set $a_t^* = k_t^*$ and $\ell_t^* = 1$, so the asset and labor markets clear.
- (2) The supporting prices satisfy the firm's first-order conditions; concavity makes them sufficient.
- (3) The planner Euler equation becomes the household Euler equation; concavity and transversality make it sufficient.
- (4) Planner feasibility is exactly goods-market clearing.

Infinite Horizon Needs a Separate Existence Argument

For every $t = 0, 1, \dots$, feasibility can be written as

$$c_t + k_{t+1} \leq k_t^\alpha + k_t.$$

This is a problem over sequences, so finite-dimensional compactness is not enough. Under bounded feasibility, continuity, discounting, and the maintained tail conditions, a limit of truncated planner problems yields a solution.

Two Separate Conclusions

planner existence + supporting prices $\implies$ equilibrium existence,
strict concavity + First Welfare Theorem $\implies$ allocation uniqueness.

Definition Comes First; Characterization Makes It Usable

1. List the primitives and every endogenous sequence.
2. Write household and firm optimization at given prices.
3. Define equilibrium by adding market clearing at every date.
4. Derive the characterization and keep all boundary conditions.
5. Use the planner problem and welfare theorems to recover allocations and supporting prices.

The Full Logic

Definition identifies the object.

Characterization reduces the search.

Theorems justify the connection.

General equilibrium is optimization made mutually consistent.